

RÓBERT LEÓ JÓNSSON

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Machine vision engineer building production robotics systems. Specialized in camera systems, 3D vision, and real-world perception.

Experience

Monumental (Machine Vision Software Engineer) *Amsterdam, Netherlands* September 2024 - present
Robotics scale-up that makes masonry robots.

- Built and deployed a photogrammetry pipeline for large-scale construction site reconstruction.
- Designed and shipped a stereo camera system for sub-mm brick analysis on a robotic end-effector.
- Developed deep learning models for 3D reconstruction and perception under harsh real-world conditions.
- Improved robot kinematics calibration and debugging workflows.

AIVA (Computer Vision and Software Engineer) *Reykjavik, Iceland* September 2023 - August 2024
Golf swing analysis using calibrated multi-camera systems

- Built real-time 3D human pose estimation from calibrated multi-camera data.
- Wrote low-level C++ for hardware-synchronized image acquisition.
- Designed and managed the PostgreSQL database.
- Designed and implemented the data pipelines for the cameras.

Aha.is (AI and Software Engineer) *Reykjavik, Iceland* April 2021 - August 2023
Demand prediction with neural networks, data analysis, and database design in MySQL.

Education

Reykjavik University August 2020 - May 2023
BSc in Discrete Mathematics and Computer Science.

- Graduated top of my class with a grade average of 9.8/10.
- Dean's list for all semesters.

Skills

- Cameras & Optics - camera/lens selection, calibration, integration
- Photogrammetry - large-scale reconstruction systems
- Stereo Vision - multi-camera systems, depth, calibration
- Deep Learning - training + deploying vision models
- Systems - real-time pipelines, concurrency, architecture

Tech

Python, C++, PyTorch, OpenCV, Ceres, Eigen, OpenGL, Nix, PostgreSQL

Personal Projects

lensboy - camera calibration library (Python) February 2026

- Developed spline-based distortion models and board warp estimation for high-precision calibration
- Designed a flexible, engineer-friendly calibration pipeline beyond standard OpenCV approaches
- Applied in real-world camera systems requiring sub-mm accuracy

slamd - GPU-accelerated 3D visualization library
Real-time 3D visualization library built with C++, OpenGL, and ImGui.

Academic Work

Expediting Self-Play Learning in AlphaZero-Style Game-Playing Agents May 2022
A reinforcement learning paper from my Bachelor's thesis. Published in the 2023 European Conference on Artificial Intelligence (ECAI).